JavaScript Regex — Main Methods

1. `test()`

Purpose: Checks whether a pattern exists. Returns `true` or `false`.

```
console.log(/\d/.test("abc123")); // true
console.log(/\d/.test("hello")); // false
```

2. `match()`

Purpose: Finds matching text in a string.

```
console.log("abc123".match(/\d+/));
// ["123"]

console.log("hello".match(/\d+/));
// null
```

3. `matchAll()`

Purpose: Finds all matches and returns their details. Requires the `g` flag.

```
let text = "a1 b2 c3";

console.log([...text.matchAll(/\d/g)]);
// Match objects for 1, 2, and 3
```

4. `replace()`

Purpose: Replaces matching text with new text.

```
console.log("abc123".replace(/\d/g, "*"));
// "abc***"

console.log("hello world".replace(/world/, "JavaScript"));
// "hello JavaScript"
```

5. `search()`

Purpose: Returns the index of the first match. Returns `-1` if no match is found.

```
console.log("abc123".search(/\d/));
// 3

console.log("hello".search(/\d/));
// -1
```

Quick Revision

- `test()` → Checks a match.
- `match()` → Finds matching text.
- `matchAll()` → Finds all matches with details.
- `replace()` → Replaces matching text.
- `search()` → Finds the index of a match.

Remember: Regex is written between forward slashes, such as `/\d+/`. The `\d` represents a digit, and `+` means one or more.